4 Partial differential equations

4.1 The heat equation

Thus far we have considered differential equations involving functions of one variable, but in reality we often have functions that depend on many variables. In this chapter we will study differential equations for functions of two variables $(x,t) \in D \times \mathbb{R}_{\geq 0}$ where $D \subseteq \mathbb{R}$. We will often use the notation

$$u_{xx}(x,t) = \frac{\partial^2 u}{\partial x^2}, \quad u_{xt}(x,t) = \frac{\partial^2 u}{\partial x \partial t}, \quad u_{tt}(x,t) = \frac{\partial^2 u}{\partial t^2}, \tag{4.1}$$

and so on to denote the partial derivatives of a function $u : D \times \mathbb{R}_{\geq 0} \to \mathbb{R}$.

The first partial differential equation (PDE) we will study is the *heat equation*. A function $u : [0,L] \times \mathbb{R}_{\geq 0} \to \mathbb{R}$ satisfies the heat equation if

$$u_t(x,t) = \kappa u_{xx}(x,t), \tag{4.2}$$

where κ is the diffusivity constant. Interpreting $u(x,t)$ as a temperature profile, κ measures how easily heat spreads throughout the domain, with large values of κ meaning that heat conducts more easily. When specifying any PDE, we must also give the boundary conditions and initial conditions in order to fully solve them. The boundary conditions of a PDE tell us about the behaviour of $u(x,t)$ at the boundaries and we will focus on two important classes of boundary conditions: *Dirichlet* and *Neumann* conditions.

Dirichlet	Neumann
$u(0,t) = \alpha,\ u(L,t) = \beta$	$u_x(0,t) = \alpha,\ u_x(L,t) = \beta$

Dirichlet conditions fix the value of $u(x,t)$ at the boundary points, while Neumann conditions allow the temperature profile to change at the boundary as long as the x-partial derivatives are maintained there. If $\alpha = \beta = 0$ in the above conditions, we say that these are *homogeneous conditions*. Homogeneous boundary conditions are nice, since if $u(x,t)$ satisfies the boundary conditions, then so does $\lambda u(x,t)$ for any $\lambda \in \mathbb{R}$. In addition to the boundary conditions, we will specify the initial conditions by fixing $u(x,0)$ as

$$u(x,0) = f(x), \quad 0 \leq x \leq L, \tag{4.3}$$

for a given function $f(x)$.

Solution with Dirichlet boundary conditions. Let us solve the heat equation with homogeneous Dirichlet boundary conditions and initial condition specified as

$$u(0,t) = u(L,t) = 0, \quad u(x,0) = f(x). \tag{4.4}$$

Generally, solutions to PDEs can display intricate dependency on both x and t, but for particularly nice PDEs we can decouple the x and t dependence and use the ansatz

$$u(x,t) = X(x)T(t). \tag{4.5}$$

This is called the *separation of variables* ansatz. Substituting this into the heat equation gives

$$X(x)T'(t) = \kappa X''(x)T(t), \tag{4.6}$$

which upon dividing by $X(x)T(t)$ gives

$$\frac{T'(t)}{T(t)} = \kappa \frac{X''(x)}{X(x)}. \tag{4.7}$$

Notice that the left-hand side of this equation only depends on t, while the right-hand side only depends on x. In order for this to be the case, we must have that both sides are actually constant! If we call this constant λ, we get two ordinary differential equations

$$T'(t) = \lambda T(t), \quad X''(x) = \frac{\lambda}{\kappa} X(x), \tag{4.8}$$

which we can easily solve. Focussing first on the equation for $X(x)$, we need to be careful about the sign of λ/κ as it will change whether exponentials or trigonometric functions will appear in the solution.

- Taking $\lambda/\kappa = \alpha^2 > 0$, we can easily solve for $X(x)$ as

  $$X(x) = A\mathrm{e}^{\alpha x} + B\mathrm{e}^{-\alpha x}. \tag{4.9}$$

  Imposing $u(0,t) = 0$ gives $A = -B$, while imposing $u(L,t) = 0$ gives

  $$0 = A(\mathrm{e}^{\alpha L} - \mathrm{e}^{-\alpha L}). \tag{4.10}$$

  Since $\mathrm{e}^{\alpha L} - \mathrm{e}^{-\alpha L} = 0$ implies that $L = 0$, we disregard this possibility and take $A = 0$. In other words, if $\lambda/\kappa > 0$ then $u(x,t) = 0$ for all values of x and t.

- If $\lambda/\kappa = 0$, then integrating $X''(x) = 0$ twice gives the solution $X(x) = Ax + B$. However, imposing the boundary conditions gives $A = B = 0$ again, so $u(x,t) = 0$ for all x and t.

- Now writing $\lambda/\kappa = -\alpha^2 < 0$, we have

  $$X(x) = A\cos\alpha x + B\sin\alpha x. \tag{4.11}$$

  The condition $u(0,t) = 0$ tells us that $A = 0$, while $u(L,t) = 0$ gives

  $$B\sin\alpha L = 0. \tag{4.12}$$

  As we're looking for non-trivial solutions, we disregard the possibility that $B = 0$, so we must impose $\sin\alpha L = 0$. This means that $\alpha L = \pi n$ for some integer n, so

  $$\alpha = \frac{n\pi}{L}, \quad n \in \mathbb{N}. \tag{4.13}$$

  Therefore, we have a solution $X(x)$ for each value of n given by

  $$X_n(x) = b_n \sin\left(\frac{n\pi x}{L}\right). \tag{4.14}$$

Having found the solutions $X_n(x)$ for $\lambda/\kappa = -\alpha^2 < 0$, we now need to find the corresponding solutions for $T(t)$. As the equation for $T(t)$ is a first order differential equation, the solution is given by

$$T_n'(t) = -\alpha^2 \kappa T_n(t) \Longrightarrow T_n(t) = C\mathrm{e}^{-\alpha^2 \kappa t}, \tag{4.15}$$

where $\alpha = n\pi/L$. Therefore, the function

$$u_n(x,t) = b_n \mathrm{e}^{-\kappa n^2 \pi^2 t/L^2} \sin\left(\frac{n\pi x}{L}\right), \tag{4.16}$$

solves the heat equation and satisfies the boundary conditions for any $n \in \mathbb{N}$. However, we've neglected the initial condition $u(x,0) = f(x)$, so this *isn't* a solution to the full problem.

In order to implement the initial condition, we note that the heat equation with homogeneous boundary conditions is *linear*, meaning that the sum of two solutions is also a solution. As such, a general solution can be written as

$$u(x,t) = \sum_{n=1}^{\infty} b_n \mathrm{e}^{-\kappa n^2 \pi^2 t/L^2} \sin\left(\frac{n\pi x}{L}\right). \tag{4.17}$$

Setting $t = 0$ and equating this to the initial configuration $f(x)$ imposes

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right). \tag{4.18}$$

This is familiar to us— it is the sine half-range series for $f(x)$! We can therefore extend $f(x)$ to an odd function on $[-L, L]$ and use the formulae from section 1.4 to find b_n as

$$b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) \mathrm{d}x. \tag{4.19}$$

In other words, solving the heat equation with homogeneous Dirichlet boundary conditions boils down to finding a Fourier sine series for the initial configuration.

Example 4.1. Suppose $u : [0,\pi] \times \mathbb{R}_{\geq 0} \to \mathbb{R}$ satisfies the heat equation with Dirichlet boundary conditions

$$u(0,t) = u(\pi,t) = 0, \tag{4.20}$$

and initial condition

$$u(x,0) = x(\pi - x). \tag{4.21}$$

As shown above, the general solution is of the form

$$u(x,t) = \sum_{n=1}^{\infty} b_n \mathrm{e}^{-\kappa n^2 t} \sin nx, \tag{4.22}$$

with coefficients given by the Fourier half-range series formula

$$\begin{aligned} b_n &= \frac{2}{\pi}\int_0^\pi x(\pi - x)\sin nx\,\mathrm{d}x, \\ &= \frac{2}{n\pi}[-x(\pi-x)\cos nx]_0^\pi + \frac{2}{n\pi}\int_0^\pi (\pi - 2x)\cos nx\,\mathrm{d}x, \\ &= \frac{2}{n^2\pi}[(\pi-2x)\sin nx]_0^\pi + \frac{4}{n^2\pi}\int_0^\pi \sin nx\,\mathrm{d}x, \\ &= \frac{4}{n^3\pi}[-\cos nx]_0^\pi = \frac{4(1-(-1)^n)}{\pi n^3}. \end{aligned} \tag{4.23}$$

When n is even, $b_n = 0$, so letting $n = 2j+1$ lets us write the solution as

$$u(x,t) = \frac{8}{\pi}\sum_{j=0}^{\infty} \frac{1}{(2j+1)^3} \mathrm{e}^{-\kappa(2j+1)^2 t}\sin(2j+1)x. \tag{4.24}$$

Solution with Neumann boundary conditions. Suppose we now impose the homogeneous Neumann boundary conditions

$$u_x(0,t) = u_x(L,t) = 0, \quad u(x,0) = f(x). \tag{4.25}$$

Of course, the heat equation itself hasn't changed, so using separation of variables $u(x,t) = X(x)T(t)$ we get the same equations

$$X''(x) = \frac{\lambda}{\kappa}X(x), \quad T'(t) = \lambda T(t). \tag{4.26}$$

Again, the behaviour of the solution depends on the value of λ/κ and we need to treat these cases separately when imposing the boundary conditions.

- For $\lambda/\kappa = \alpha^2 > 0$, the solution is given by a sum of exponentials

  $$X(x) = A\mathrm{e}^{\alpha x} + B\mathrm{e}^{-\alpha x}. \tag{4.27}$$

  As the Neumann conditions involve the derivatives $u_x(x,t) = X'(x)T(t)$, we impose $X'(0) = X'(L) = 0$ where

  $$X'(x) = A\alpha\mathrm{e}^{\alpha x} - B\alpha\mathrm{e}^{-\alpha x}. \tag{4.28}$$

  Substituting in $x = 0$ and $x = L$ readily shows that $A = B = 0$, giving $u(x,t) = 0$ for all x and t.

- When $\lambda/\kappa = 0$ we get $X(x) = Ax + B$ and $X'(x) = A$. Imposing the boundary conditions gives $A = 0$, but doesn't fix B. Therefore, we get a possible solution with $X(x)$ constant. However, since the general solution will not depend on x at all, we can only have this solution if $u(x,0)$ is also constant.

- Finally, when $\lambda/\kappa = -\alpha^2 < 0$ we obtain the trigonometric solution

$$X(x) = A\cos\alpha x + B\sin\alpha x \Longrightarrow X'(x) = -A\alpha\sin\alpha x + B\alpha\cos\alpha x. \tag{4.29}$$

The condition $u_x(0,t) = 0$ tells us that $B = 0$, while $u_x(L,t) = 0$ gives

$$A\alpha\sin\alpha L = 0 \Longrightarrow \alpha = \frac{n\pi}{L}, \tag{4.30}$$

for $n \in \mathbb{Z}_{\geq 0}$. Therefore, we get the solutions

$$X_n(x) = a_n\cos\left(\frac{n\pi x}{L}\right). \tag{4.31}$$

Thus far, the only difference to the Dirichlet case is that we obtain solutions with cosines instead of sines.

Solving for $T(t)$ is exactly the same as before, therefore the general solution takes the form

$$u(x,t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \mathrm{e}^{-\kappa n^2\pi^2 t/L^2}\cos\left(\frac{n\pi x}{L}\right) \tag{4.32}$$

Imposing the initial condition $u(x,0) = f(x)$ now gives

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos\left(\frac{n\pi x}{L}\right), \tag{4.33}$$

and we recognise this as the cosine half-range series for $f(x)$. As in the Dirichlet case, we obtain the expressions for a_n by using the half-range formulae

$$a_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)\mathrm{d}x. \tag{4.34}$$

Overall, in both the Dirichlet and Neumann case we obtain a series expansion that depends exponentially on t but trigonometrically on x with the only difference being the type of Fourier half-range series that appears for each type of boundary condition. Therefore, for the heat equation we have the motto that 'Dirichlet = sine series' and 'Neumann = cosine series'.

Example 4.2. Suppose $u : [0,1]\times\mathbb{R}_{\geq 0} \to \mathbb{R}$ satisfies the heat equation with Neumann boundary conditions

$$u_x(0,t) = u_x(1,t) = 0, \tag{4.35}$$

and initial condition

$$u(x,0) = x. \tag{4.36}$$

The general solution takes the form

$$u(x,t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \mathrm{e}^{-\kappa n^2\pi^2 t}\cos(n\pi x), \tag{4.37}$$

where the coefficients for $n > 0$ are given by

$$\begin{aligned} a_n &= 2\int_0^1 x\cos(n\pi x)\,\mathrm{d}x, \\ &= \frac{2}{n\pi}[x\sin(n\pi x)]_0^1 - \frac{2}{n\pi}\int_0^1 \sin(n\pi x)\,\mathrm{d}x, \\ &= \frac{2}{(n\pi)^2}[\cos(n\pi x)]_0^1 = \frac{2((-1)^n-1)}{(n\pi)^2}, \end{aligned} \tag{4.38}$$

which is zero when n is even. Since this is singular when $n = 0$, we need to find a_0 separately. Explicitly, we have

$$a_0 = 2\int_0^1 x\,\mathrm{d}x = 1. \tag{4.39}$$

Writing $n = 2j + 1$, the solution is given by

$$u(x,t) = \frac{1}{2} - \frac{4}{\pi^2}\sum_{j=0}^{\infty}\frac{1}{(2j+1)^2}\mathrm{e}^{-\kappa(2j+1)^2\pi^2 t}\cos(2j+1)\pi x. \tag{4.40}$$

4.2 The wave equation

The second PDE we will deal with is the *wave equation* that describes the propagation of waves through a medium. We'll focus on the 1-dimensional case where the function $u : [0, L]\times\mathbb{R}_{\geq 0} \to \mathbb{R}$ satisfies

$$u_{tt}(x,t) = c^2 u_{xx}(x,t). \tag{4.41}$$

Here we can think of $u(x,t)$ as the height of a wave over a point x at a time t and the constant c describes the propagation speed of the wave. We will also imagine that the wave is fixed at the boundary of the interval $[0, L]$ at the same level, therefore restricting us to homogeneous Dirichlet boundary conditions

$$u(0,t) = u(L,t) = 0. \tag{4.42}$$

Additionally, as there is an additional t-derivative when compared with the heat equation we need more initial conditions. Therefore, we must specify

$$u(x,0) = f(x), \quad u_t(x,0) = g(x). \tag{4.43}$$

In other words, we need to specify both the initial profile $f(x)$ of the wave along with its initial velocity profile $g(x)$.

We proceed as before by using separation of variables to write $u(x,t) = X(x)T(t)$, giving

$$X(x)T''(t) = c^2 X''(x)T(t), \tag{4.44}$$

which, upon division by $X(x)T(t)$, we get

$$\frac{T''(t)}{T(t)} = c^2 \frac{X''(x)}{X(x)}. \tag{4.45}$$

Since the left-hand side depends only on t, while the right-hand side depends only on x, these must be equal to a constant λ. Hence

$$T''(t) = \lambda T(t), \quad X''(x) = \frac{\lambda}{c^2} X(x). \tag{4.46}$$

The only difference when compared to the heat equation is that the equation for $T(t)$ is a second order differential equation, rather than first order. However, solving for $X(x)$ is completely analogous to the heat equation: we consider the cases $\lambda/c^2 < 0$, $\lambda = 0$ and $\lambda/c^2 > 0$ separately and come to the conclusion that only the $\lambda/c^2 < 0$ case is important. Restricting to this case, we get

$$X(x) = A\cos\alpha x + B\sin\alpha x, \tag{4.47}$$

where $\lambda/c^2 = -\alpha^2$. The boundary condition $u(0,t) = 0$ gives us $A = 0$, while $u(L,t) = 0$ constrains α to

$$\alpha = \frac{n\pi}{L}, \quad n \in \mathbb{N}. \tag{4.48}$$

Fixing the value of α, the corresponding solution for $T(t)$ is then

$$T_n(t) = C_n \cos\left(\frac{n\pi ct}{L}\right) + D_n \sin\left(\frac{n\pi ct}{L}\right). \tag{4.49}$$

Unlike the heat equation, we now get trigonometric dependence on both t and x. The wave equation with homogeneous boundary conditions is linear, so we can sum our solutions together as before. This gives the general solution

$$u(x,t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right)\left(C_n \cos\left(\frac{n\pi ct}{L}\right) + D_n \sin\left(\frac{n\pi ct}{L}\right)\right). \tag{4.50}$$

Finally, we need to apply the initial conditions, both to $u(x,0)$ and the velocity profile $u_t(x,0)$. Using the initial condition $u(x,0) = f(x)$, we have

$$f(x) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi x}{L}\right), \tag{4.51}$$

realising $f(x)$ as a Fourier sine series. Therefore, the coefficients C_n are

$$C_n = \frac{2}{L}\int_0^L f(x)\sin\left(\frac{n\pi x}{L}\right)\mathrm{d}x. \tag{4.52}$$

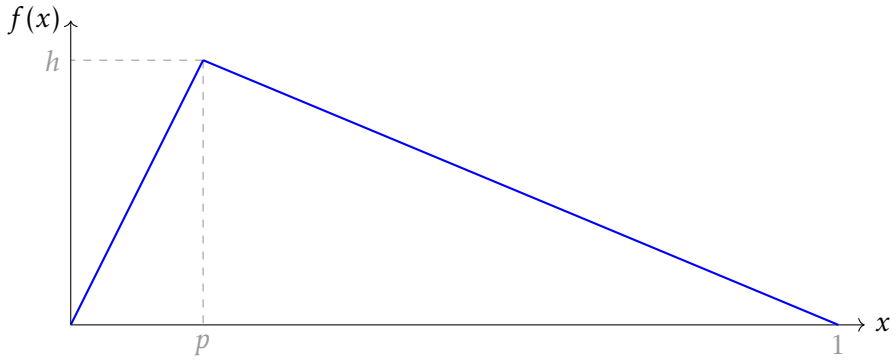


Figure 8: The initial configuration of the plucked string.

Applying the initial condition $u_t(x,0)$ gives

$$g(x) = \sum_{n=1}^{\infty} D_n \left(\frac{n\pi c}{L}\right) \sin\left(\frac{n\pi x}{L}\right). \tag{4.53}$$

Again, this is a Fourier sine series, but we should be careful when using the coefficient formula to include the $n\pi c/L$ factor. Doing so gives

$$D_n\left(\frac{n\pi c}{L}\right) = \frac{2}{L}\int_0^L g(x)\sin\left(\frac{n\pi x}{L}\right)\mathrm{d}x, \tag{4.54}$$

and dividing by $n\pi c/L$ gives the coefficients

$$D_n = \frac{2}{n\pi c}\int_0^L g(x)\sin\left(\frac{n\pi x}{L}\right)\mathrm{d}x. \tag{4.55}$$

All in all, solving the wave equation with homogeneous Dirichlet boundary conditions reduces to finding the Fourier sine series for the initial configuration $f(x)$ and the initial velocity profile $g(x)$, with an additional factor $L/n\pi c$ included in the coefficients for the latter.

Example 4.3. We want to model a string that is plucked to a height h at some point p along its length. By taking the string to lie on the interval $[0,1]$, the initial configuration is given by (drawn in fig. 8)

$$f(x) = \begin{cases} hx/p, & 0 \le x \le p, \\ h(1-x)/(1-p), & p < x \le 1. \end{cases} \tag{4.56}$$

We will assume that the string is at rest as it is plucked, so $u_t(x,0) = 0$. Using this in the equation for the D_n coefficients, we see that $D_n = 0$ for every n. The solution is therefore of the form

$$u(x,t) = \sum_{n=1}^{\infty} C_n \sin(n\pi x)\cos(n\pi c t). \tag{4.57}$$

Now we use the Fourier sine series formula to get

$$C_n = 2\int_0^1 f(x)\sin(n\pi x)\,\mathrm{d}x = \frac{2h}{p}\int_0^p x\sin(n\pi x)\,\mathrm{d}x + \frac{2h}{1-p}\int_p^1 (1-x)\sin(n\pi x)\,\mathrm{d}x. \tag{4.58}$$

Call the first integral on the right-hand side I_1 and the second I_2. Using integration by parts, we calculate I_1 as

$$\begin{aligned} I_1 &= \frac{2h}{p}\int_0^p x\sin(n\pi x)\,\mathrm{d}x, \\ &= \frac{2h}{n\pi p}\left[-x\cos(n\pi x)\right]_0^p + \frac{2h}{n\pi p}\int_0^p \cos(n\pi x)\,\mathrm{d}x, \\ &= -\frac{2h}{n\pi}\cos(n\pi p) + \frac{2h}{(n\pi)^2 p}\left[\sin(n\pi x)\right]_0^p = \frac{2h}{(n\pi)^2 p}\sin(n\pi p) - \frac{2h}{n\pi}\cos(n\pi p). \end{aligned} \tag{4.59}$$

Similarly, integration by parts on I_2 gives

$$\begin{aligned} I_2 &= \frac{2h}{1-p}\int_p^1 (1-x)\sin(n\pi x)\,\mathrm{d}x, \\ &= \frac{2h}{n\pi(1-p)}\left[-(1-x)\cos(n\pi x)\right]_p^1 - \frac{2h}{n\pi(1-p)}\int_p^1 \cos(n\pi x)\,\mathrm{d}x, \\ &= \frac{2h}{n\pi}\cos(n\pi p) - \frac{2h}{(n\pi)^2(1-p)}\left[\sin(n\pi x)\right]_p^1, \\ &= \frac{2h}{(n\pi)^2(1-p)}\sin(n\pi p) + \frac{2h}{n\pi}\cos(n\pi p). \end{aligned} \tag{4.60}$$

When we add I_1 and I_2 the cosine term cancels, so we arrive at

$$\begin{aligned} C_n &= \frac{2h}{(n\pi)^2 p}\sin(n\pi p) + \frac{2h}{(n\pi)^2(1-p)}\sin(n\pi p), \\ &= \frac{2h\sin(n\pi p)}{p(1-p)(n\pi)^2}. \end{aligned} \tag{4.61}$$

Now that we've done the painful work, we have that the solution to the wave equation with the plucked string initial conditions is given by

$$u(x,t) = \frac{2h}{p(1-p)\pi^2}\sum_{n=1}^{\infty}\left(\frac{\sin(n\pi p)}{n^2}\right)\sin(n\pi x)\cos(n\pi c t). \tag{4.62}$$

Choosing some sensible values for h, c and p and plotting this for various values of t shows that the wave propagates away from the plucked point until it hits a boundary, thereafter it is reflected back towards the other direction. Compare this with this slow motion video of a plucked string.

4.3 Solving inhomogeneous PDEs

Suppose we are given a partial differential equation together with some boundary conditions. We say that it is *inhomogeneous* if there exist solutions $u(x,t)$ and $v(x,t)$ satisfying the boundary conditions such that a linear combination $\alpha u(x,t) + \beta v(x,t)$ does not satisfy the PDE with specified boundary conditions. For example, consider the *forced heat equation*

$$u_t(x,t) - \kappa u_{xx}(x,t) = f(x), \tag{4.63}$$

satisfying $u(0,t) = u(L,t) = 0$. If $u(x,t)$ and $v(x,t)$ are solutions, then $u(x,t) + v(x,t)$ still satisfies the boundary conditions, but we have that

$$\frac{\partial}{\partial t}(u+v) - \kappa \frac{\partial^2}{\partial x^2}(u+v) = (u_t - \kappa u_{xx}) + (v_t - \kappa v_{xx}). \tag{4.64}$$

Each of the bracketed terms is just the left-hand side of eq. (4.63), so we have

$$(u+v)_t - \kappa(u+v)_{xx} = 2f(x). \tag{4.65}$$

Therefore, $u(x,t)+v(x,t)$ is ***not*** a solution to the PDE. Generally, given a linear PDE, adding a term independent of the solution $u(x,t)$ breaks the homogeneity of the system. Here the forcing term $f(x)$ is responsible for this.

A second way homogeneity can fail is through its boundary conditions. If we now consider the heat equation with boundary conditions $u(0,t) = u(L,t) = \pi$, then the sum of two solutions $u(x,t)$ and $v(x,t)$ satisfies the boundary conditions

$$(u+v)(0,t) = (u+v)(L,t) = 2\pi. \tag{4.66}$$

These are the wrong boundary conditions! For Dirichlet and Neumann boundary conditions, the system can be homogeneous only if $u(0,t) = u(L,t) = 0$ or $u_x(0,t) = u_x(L,t) = 0$.

So how can we solve eq. (4.63) with general inhomogeneous boundary conditions $u(0,t) = \alpha$ and $u(L,t) = \beta$? We can do so by writing $u(x,t)$ as

$$u(x,t) = v(x,t) + w(x), \tag{4.67}$$

where $v(x,t)$ satisfies the homogeneous heat equation

$$v_t(x,t) - \kappa v_{xx}(x,t) = 0, \tag{4.68}$$

with boundary conditions $v(0,t) = v(L,t) = 0$. The function $w(x)$ is called a ***stationary solution***, as it does not depend on t. Its purpose is to adjust the homogeneous solution so we can satisfy both the forcing and the boundary conditions. Substituting $u(x,t)$ into the left-hand side of the forced heat equation then gives

$$(v(x,t) + w(x))_t - \kappa(v(x,t) + w(x))_{xx} = (v_t(x,t) - \kappa v_{xx}(x,t)) - \kappa w''(x). \tag{4.69}$$

By hypothesis, the term in the bracket vanishes, so we get the condition

$$w''(x) = -\frac{f(x)}{\kappa}. \tag{4.70}$$

Imposing the inhomogeneous boundary conditions also implies that

$$w(0) = \alpha, \quad w(L) = \beta. \tag{4.71}$$

Therefore, in order to solve the forced heat equation, we need to solve the second order ODE eq. (4.70) subject to the boundary conditions above. The only caveat occurs when considering the initial condition

$$u(x,0) = g(x) \Longrightarrow v(x,0) = g(x) - w(x). \tag{4.72}$$

Therefore, the full solution is given by

$$u(x,t) = w(x) + \sum_{n=1}^{\infty} b_n \mathrm{e}^{-\kappa n^2 \pi^2 t/L^2} \sin\left(\frac{\pi n x}{L}\right), \tag{4.73}$$

where the coefficients are given by

$$b_n = \frac{2}{L}\int_0^L (g(x) - w(x)) \sin\left(\frac{\pi n x}{L}\right) \mathrm{d}x. \tag{4.74}$$

This idea generalises to other inhomogeneous PDEs: split up $u(x,t)$ as $v(x,t)+w(x)$ with $v(x,t)$ satisfying the homogeneous version of the problem, substitute this into the equation to find an ODE for $w(x)$ and solve it. Then rewrite the initial condition for $v(x,t)$ and calculate the appropriate Fourier coefficients.

Example 4.4. For the first example, consider the unforced heat equation

$$u_t(x,t) = \kappa u_{xx}(x,t), \tag{4.75}$$

with the boundary and initial conditions

$$u(0,t) = 0, \quad u(\pi,t) = -\pi^2, \quad u(x,0) = -x^2. \tag{4.76}$$

First, write $u(x,t) = v(x,t) + w(x)$ with $v(x,t)$ the solution to the homogeneous problem and $w(x)$ a stationary solution. Substituting this into the heat equation yields

$$w''(x) = 0 \Longrightarrow w(x) = Ax + B, \tag{4.77}$$

where A and B are constants of integration. The boundary condition $u(0,t) = 0$ implies $B = 0$, while the other boundary condition gives

$$A\pi = -\pi^2 \Longrightarrow w(x) = -\pi x. \tag{4.78}$$

The solution is therefore of the form

$$u(x,t) = -\pi x + \sum_{n=1}^{\infty} b_n \mathrm{e}^{-\kappa n^2 t} \sin nx. \tag{4.79}$$

The coefficients can be found by using the initial condition for $v(x,t)$ given by

$$v(x,0) = u(x,0) - w(x) = \pi x - x^2 = x(\pi - x), \tag{4.80}$$

hence

$$b_n = \frac{2}{\pi}\int_0^\pi x(\pi - x)\sin nx\,\mathrm{d}x. \tag{4.81}$$

Thankfully, we calculated these integrals in example 4.1, so we can conclude that the solution is

$$u(x,t) = -\pi x + \frac{8}{\pi}\sum_{j=0}^{\infty}\frac{1}{(2j+1)^3}\mathrm{e}^{-\kappa(2j+1)^2 t}\sin(2j+1)x. \tag{4.82}$$

Example 4.5. Consider the forced wave equation

$$u_{tt}(x,t) - u_{xx}(x,t) = \sin\pi x, \tag{4.83}$$

with boundary and initial conditions

$$u(0,t) = u(1,t) = 0, \quad u(x,0) = 0, \quad u_t(x,0) = \sin 3\pi x. \tag{4.84}$$

Decompose the solution as $u(x,t) = v(x,t) + w(x)$ with $v(x,t)$ the homogeneous solution. Substituting this into the forced wave equation gives

$$w''(x) = -\sin\pi x. \tag{4.85}$$

Integrating this twice gives the stationary solution

$$w(x) = \frac{1}{\pi^2}\sin\pi x + Ax + B \tag{4.86}$$

Since $\sin 0 = \sin\pi = 0$, the boundary conditions impose $A = B = 0$. Therefore, the solution takes the form

$$u(x,t) = \frac{1}{\pi^2}\sin\pi x + \sum_{n=1}^{\infty}\sin(n\pi x)\left(C_n\cos(n\pi t) + D_n\sin(n\pi t)\right). \tag{4.87}$$

The initial condition for $v(x,t)$ is $v(x,0) = -w(x)$, so the C_n coefficients are given by

$$C_n = -\frac{2}{\pi^2}\int_0^1 \sin\pi x\sin n\pi x\,\mathrm{d}x = \begin{cases} -1/\pi^2, & n = 1, \\ 0, & \text{otherwise.} \end{cases} \tag{4.88}$$

This can be justified by using the product formula for sines or by expanding each term into complex exponentials. As $w(x)$ does not depend on t, the initial condition for $v_t(x,t)$ is the

same as $u_t(x,0) = \sin 3\pi x$. As such, the coefficient D_n is

$$D_n = \frac{2}{n\pi}\int_0^1 \sin 3\pi x \sin n\pi x \,\mathrm{d}x = \begin{cases} 1/3\pi, & n = 3, \\ 0, & \text{otherwise.} \end{cases} \tag{4.89}$$

Therefore, only C_1 and D_3 are non-zero. As such, the full solution is given by

$$u(x,t) = \frac{1}{\pi^2}\sin \pi x - \frac{1}{\pi^2}\sin \pi x \cos \pi t + \frac{1}{3\pi}\sin 3\pi x \sin 3\pi t. \tag{4.90}$$

4.4 The heat equation on an infinite domain ★

Just as Fourier series can be used to solve the heat equation on an interval $[0, L]$, the Fourier transform can help us solve it over the whole real line $\mathbb{R}$. Concretely, suppose $u : \mathbb{R} \times \mathbb{R}_{\geq 0} \to \mathbb{R}$ satisfies the heat equation with initial condition

$$u(x,0) = f(x), \tag{4.91}$$

where $f \in L^1(\mathbb{R})$. We will assume that $u(x,t) \in L^1(\mathbb{R})$ for any fixed value of t, so $\lim_{x\to\pm\infty} u(x,t) = 0$. By assuming this, we can define the Fourier transform of $u(x,t)$ with respect to the x variable as usual

$$\hat{u}(\xi,t) = \int_{-\infty}^{\infty} u(x,t) \mathrm{e}^{-\mathrm{i}\xi x}\,\mathrm{d}x. \tag{4.92}$$

Taking the Fourier transform of the heat equation then gives

$$\hat{u}_t(\xi,t) = -\kappa\xi^2 \hat{u}(\xi,t). \tag{4.93}$$

This is a first order differential equation for $\hat{u}(\xi,t)$ that we can solve easily by rearranging and integrating as

$$\int \frac{1}{\hat{u}}\frac{\partial \hat{u}}{\partial t}\,\mathrm{d}t = -\kappa \int \xi^2 \,\mathrm{d}t \implies \log \hat{u}(\xi,t) = C(\xi) - \kappa\xi^2 t. \tag{4.94}$$

Here the constant of integration $C(\xi)$ can depend on ξ since we've integrated with respect to t. Defining $D(\xi) = \mathrm{e}^{C(\xi)}$, we can write the solution as

$$\hat{u}(\xi,t) = D(\xi)\mathrm{e}^{-\kappa\xi^2 t}. \tag{4.95}$$

The initial condition can also be Fourier transformed as $\hat{u}(\xi,0) = \hat{f}(\xi)$, hence

$$\hat{f}(\xi) = D(\xi). \tag{4.96}$$

Piecing this all together and using the Fourier inversion theorem brings us to the full solution

$$u(x,t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \hat{f}(\xi)\mathrm{e}^{\mathrm{i}\xi x - \kappa\xi^2 t}\,\mathrm{d}\xi. \tag{4.97}$$

Since this is an integral over ξ, one can check that this satisfies the heat equation by differentiating with respect to x and t under the integral sign.

As a simple example, consider the initial configuration $u(x,0) = \delta(x-y)$ for some point $y \in \mathbb{R}$. Then the Fourier transformed initial condition is

$$\hat{u}(\xi,0) = \int_{-\infty}^{\infty} \delta(x-y) \mathrm{e}^{-\mathrm{i}\xi x} \,\mathrm{d}x = \mathrm{e}^{-\mathrm{i}\xi y}, \tag{4.98}$$

and the full solution is

$$u(x,t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}\xi(x-y)-\kappa\xi^2 t} \,\mathrm{d}\xi = \mathcal{F}^{-1}\left[\mathrm{e}^{-\mathrm{i}\xi y-\kappa\xi^2 t}\right]. \tag{4.99}$$

By using the transformation rules for Fourier transforms, we can deduce that

$$u(x,t) = \frac{1}{\sqrt{4\pi\kappa t}} \exp\left(-\frac{(x-y)^2}{4\kappa t}\right). \tag{4.100}$$

This is known as the *heat kernel*, a useful fundamental solution to the heat equation which has found usage in many fields of mathematics.